Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/05/2015
PI Name:	Stamper-Kurn, Dan Ph.D.		
Project Title:	Coherent Magnon Optics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dmsk@berkeley.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	510-642-9618
Organization Name:	University of California, Berkeley		
PI Address 1:	Physics		
PI Address 2:	366 Leconte Hall 7300		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-7301	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	04/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mueller, Holger Ph.D. (University of California, Berkeley)		
Grant/Contract No.:	JPL 1505588		
Performance Goal No.:			
Performance Goal Text:			
	The spin degree of freedom of ultracold atoms makes the physics of quantum gases richer, and offers opportunities for fundamental science explorations using cold atomic media. The scientific agenda we propose for the Cold Atom Laboratory (CAL) builds upon on our world-leading research on spinor Bose-Einstein condensates, including our development of magnetization-sensitive imaging and studies of spin coherence in optically trapped quantum gases. We propose four activities: (1) Precise characterization of magnons in spinor Bose-Einstein condensates: Magnons are low energy spin excitations in spinor condensates. Theories of symmetry breaking, as well as mean-field and beyond-mean-field mean-field treatments of Bose gases provide strong predictions that we will test precisely with interferometric measurements based		

Task Description:	on coherent magnon optics. (2) Magnon and Bragg interferometry in dense and dilute Bose condensed gases: magnons in ferromagnetic spinor condensates, such as the $F=1$ gases of 87Rb and 41K, are by necessity gapless and quadratically dispersing, with an effective mass that, within mean field theory that is applicable to weakly interacting gases, is equal to the atomic mass. In other words, magnons propagate just like free particles. We propose to implement an atom interferometer based on coherent magnon optics. Compared to a Bragg interferometer, a magnon interferometer has no mean-field interaction shift. Comparing the two interferometers provides a strong measurement cross-check and may allow quantitative studies of polaronic (quantum depletion) energy shifts. (3) Imaging superflow with a magnon shear interferometer: We propose to image the magnon interference pattern to perform in-situ measurements of superfluid flow in three dimensions, akin to particle tracking velocimetry. We propose to utilize the CAL to expand a trapped Bose-Einstein condensate into a large three dimensional volume, essentially magnifying the superfluid flow pattern sufficiently that it can be imaged. A three-dimensional spatial pattern of magnons will be allowed to propagate coherently before being imaged, revealing the three-dimensional superfluid velocity field. (4) Magnon-evaporative cooling: We will explore a novel cooling technique in which magnon excitations are used to absorb, and then expel entropy from a highly degenerate spinor Bose-Einstein condensate. Our target is to produce quantum gases at the lowest entropy per particle, with a method that can be applied in the CAL apparatus.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is fundamental in flavor. We are exploring new phenomena and new experimental capabilities in a physical system that is itself new and only recently available for experimentation. Exploring in this new space of questions has prospects for making discoveries. More specifically, the research impacts our knowledge of materials (through studies of superfluid turbulence), extends the frontiers of natural phenomena that we can access and explore (through the development of new thermometry and cooling techniques), and contributes to the development of high precision sensors for future scientific and technological applications (through studies of atom and magnon interferometry).
Task Progress:	During this first year of NASA support, we have made significant progress on two of the specific aims of this grant: the precise characterization of magnons, and the implementation of magnon cooling to a ferromagnetic atomic gas. We completed our first study of the properties of magnon excitations in a spinor Bose-Einstein condensed gas, these being the low-energy magnetic excitations of a system that is both ferromagnetic and also superfluid. We developed an optical method for producing coherent waves of magnons and then a spin-specific imaging method that allowed us to visualize the evolution of these waves. These methods were used to construct a magnon-wave interferometer that was suited to measuring the magnon recoil energy -- the extra kinetic energy that a magnon acquires as a function of its wavelength or velocity. We were testing theoretical predictions that the recoil energy should scale quadratically with the wavevector of the magnon and that the effective mass of a magnon should be very close to (with about 0.3%) the mass of a bare rubidium atom. Our data supported the prediction of a quadratic dispersion relation, but identified the magnon mass to be significantly higher than was predicted. We do not understand the reason for this heavy mass yet. In addition, we measured the magnon energy gap -- the energy required to create a stationary magnon. Applying the cold atomic gas as a magnetic field sensor, we measured the field produced by the gas itself and identified this field as the origin for the magnon energy gap. The gap was measured to be around 1 Hertz (in units of frequency), in good agreement with predictions. We also made significant progress on the project's aim of developing techniques to measure and to lower the temperature of ultracold trapped atomic gases. Here, we aim to overcome limitations of current schemes of thermometry and cooling, both of which require one to identify the rare thermally excited atoms that exist in a highly degenerate atomic gas. We found that introducing an incoherent population of magnon excitations to the gas allowed one to measure the temperature in regimes that were hitherto inaccessible to measurement. From such measurements, we found that atomic Bose gases can be produced at extremely low values of the entropy per particle, a finding that bodes well for experiments that seek to model low-temperature materials using gases of cold atoms. In addition, we found that the very act of producing this incoherent magnon gas, out of an initially coherent wave of magnons, tended to lower the temperature of the gas. We identified this as a form of isoenergetic demagnetization cooling, and demonstrated it to be efficient in reducing the temperature and entropy of gases trapped in deep traps. Both the thermometry and the magnon cooling methods can be adapted in a straightforward manner to experiments being planned aboard the Cold Atom Laboratory on the International Space Station. Recently, we have begun a new effort to characterize the re-condensation of magnons within a spinor Bose-Einstein condensate. The project has just gotten underway, but we are already seeing key features that are expected of this magnon condensation transition, such as the dependence of the critical atom number on system parameters, and the spontaneous symmetry breaking that occurs at such condensation.
Bibliography Type:	Description: (Last Updated: 06/16/2021)
Articles in Peer-reviewed Journals	Marti GE, MacRae A, Olf R, Lourette S, Fang F, Stamper-Kurn DM. "Coherent magnon optics in a ferromagnetic spinor Bose-Einstein condensate." Physical Review Letters. 2014 Oct;113:155302. http://dx.doi.org/10.1103/PhysRevLett.113.155302 , Oct-2014